

HDOC DAY_22 PROGRAMMING IN C

Problem

Write a menu-driven C program that accepts two positive integers, m and n , where $m \leq n$. Display the following menu and use a switch statement to perform the operation selected by the user:

1. Print the multiplication tables of m and n side by side, using multipliers from 1 to 10.
2. Display a multiplication grid containing the products of integers from 1 to m and integers from 1 to n .
3. For each integer from m to n , print that integer followed by the next five consecutive integers.
4. Exit the program.

Continue displaying the menu until the user selects the Exit option. Display an appropriate error message if the user enters an invalid menu choice.

ALGORITHM

? Step 1: Start

? Step 2: Initialize Control Variable

Set loop control variable $z = 1$.

? Step 3: Main Program Loop

While $z == 1$, repeat **Steps 4 through 9**:

? Step 4: Input Numbers

Prompt the user to enter two positive integers: n (first integer, intended to be larger) and m (second integer, intended to be smaller). Read n and m .

? Step 5: Display Menu & Get Choice

Print the available menu options:

1. Print side-by-side multiplication tables of m and n .
2. Display a multiplication grid from 1 to m (rows) and 1 to n (columns).
3. Print consecutive sequence ranges from m to n .
4. Exit program.

Read the user's menu choice into variable ch.

? Step 6: Execute Choice (Switch Statement)

- **Case 1: Side-by-Side Multiplication Tables**

- For counter variable i from 0 to 10 (inclusive):
 - Print $m * i = (m * i)$ and $n * i = (n * i)$ side by side separated by tab spaces.
- Print a new line.

- **Case 2: Multiplication Grid**

- **Header:** Print leading spaces, then for i from 1 to n, print i as column headers. Print a new line.
- **Rows:** For row counter i from 1 to m:
 - Print row label i followed by |.
 - For column counter j from 1 to n:
 - Print the product $i * j$.
 - Print a new line.

- **Case 3: Consecutive Integer Sequences**

- For outer loop i from m to n:
 - For inner loop j from 0 to 5:
 - Print $i:(i + j),.$
 - Print a new line.

- **Case 4: Exit**

- Set $z = 0$ to terminate the loop.

- **Default (Invalid Option)**

- Display message: "invalid choice".

? Step 7: Loop Check

Return to **Step 3**. If $z == 1$, run again; otherwise, break loop.

? Step 8: End

Return 0 and terminate execution.

C PROGRAM CODE

```
#include <stdio.h>

int main()
{
    int z=1;
    while(z==1)
    { int n,m,ch;

        printf("enter 2 positive numbers 1st greater and second smaller");

        scanf("%d%d",&n,&m);

        printf("\nchoice 1 is to Print the multiplication tables of m and n side by side, using
multipliers from 1 to 10.");

        printf("\nchoice 2 is to Display a multiplication grid containing the products of
integers from 1 to m and integers from 1 to n.");

        printf("\nchoice 3 For each integer from m to n, print that integer followed by the
next five consecutive integers.");

        printf("\nchoice 4 is to exit");

        printf("enter your choice\n");

        scanf("%d",&ch);

        switch (ch)
        {
            case 1:
            {
                for(int i=0;i<=10;i++)
                {
                    printf("%d*%d=%d\t%d*%d=%d",m,i,(m*i),n,i,(n*i));

                }

                printf("\n");

                break;
            }
        }
    }
```

case 2:

```
{ printf("  ");
  for(int i=1;i<=n;i++)
  {
    printf("\t%d\t",i);
  }printf("\n");
  for(int i=1;i<=m;i++)
  { printf("%d |",i);
    for(int j=1;j<=n;j++)
    {
      printf("\t%d\t",(i*j));
    }
    printf("\n");
  }
  break;
}
```

case 3:

```
{
  for(int i=m;i<=n;i++)
  {
    for(int j=0;j<=5;j++)
    {
      printf("%d:%d",i,(j+i));
    }
    printf("\n");
  }
  break;
}
```

```
    case 4:
    {
        z = 0;
        break;
    }
    default:
    {
        printf("invalid choice");
        break;
    }
}
return 0;
}
```